Elliptic PDEs

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Lecture 1

0 Introduction

We will study second order elliptic PDEs on (a domain in) $\mathbb{R}^n$. For example, arising from variational problems. Ultimately, we are interested in non-linear PDEs. To do this, we will first understand the linear theory.

Setup: Consider a domain $\Omega \subseteq \mathbb{R}^n$, i.e. open, bounded and connected, and a function

$$\begin{aligned} F : \Omega \times \mathbb{R} \times \mathbb{R}^n &\to \mathbb{R} \\ (x, z, p) &\mapsto F(x, z, p) \end{aligned}$$

and consider the functional

$$\mathcal{F}[u] = \int_\Omega F(x, u(x), \partial u(x)) \mathrm{d}x$$

Note we will use ∂, D, ∇ essentially interchangably. Assume F is sufficiently regular. Let $u \in \mathcal{S}$, a suitable vector space of functions $u : \Omega \to \mathbb{R}$. Frequently, $\mathcal{S} = \mathrm{H}^1(\Omega)$, or $\mathcal{S} = C^{1,\alpha}(\Omega)$.

Suppose u minimises $\mathcal{F}$, subject to $u|_{\partial\Omega} = g$, for some given $g : \partial\Omega \to \mathbb{R}$[1]. So for all $\varphi \in \mathcal{S}$,

$$\mathcal{F}[u + t\varphi] \geq \mathcal{F}[u]$$

In particular, this means that

$$\left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=0} \mathcal{F}[u + t\varphi] = 0$$

Or another words,

$$\left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=0} \int_\Omega F(x, u + t\varphi, \partial u + t\partial\varphi) \mathrm{d}x = 0$$

Assume enough regularity so that we can exchange the derivative and integral, we get that

$$\int_\Omega (\partial_z F)(x, u, \partial u)\varphi + \partial_i \varphi (\partial_{p_i} F)(x, u, \partial u) \mathrm{d}x = 0 \tag{1}$$

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[1] Boundary conditions are needed for well-posedness

As usual, we will use the summation convention. To ensure that $u + t\varphi$ satisfies the correct boundary conditions, $\varphi|_{\partial\Omega} = 0$. Integrate eq. (1) by parts, we get

$$\int_\Omega \varphi(x) \left(\partial_z F - \partial_i \partial_{p_i} \mathcal{F}\right)(x, u, \partial u) \mathrm{d}x = 0$$

This is true for all $\varphi \in \mathcal{S}$, and so by the fundamental lemma of calculus of variations,

$$\frac{\partial F}{\partial z} - \frac{\partial}{\partial x_i}\left(\frac{\partial F}{\partial p_i}\right) = 0$$

which is the Euler-Lagrange equation for F. We can rewrite this as

$$\frac{\partial F}{\partial z} - \partial_i \partial_j u \frac{\partial^2 F}{\partial p_i \partial p_j} = 0 \tag{2}$$

This is now a second order quasilinear PDE in u. More generally, consider

$$a^{ij}(x, u, \partial u)\partial^2_{ij} u - b(x, u, \partial u) = 0 \tag{3}$$

Definition 0.1
We say that eq. (3) is *elliptic* in Ω if $a^{ij}(x, u, \partial u)$ is a positive definite matrix in Ω.

In the case of eq. (2), this is equivalent to F being convex in the variable p.

Example 0.2 (Dirichlet energy)
When

$$F(x, z, p) = |p|^2$$

One gets

$$\Delta u = 0$$

Extremisers of this are called *harmonic functions*.

Example 0.3 (Minimal surfaces)
When

$$F(x, z, p) = \sqrt{1 + |p|^2}$$

We leave as an exercise to interpret $\mathcal{F}[u]$. In this case, we get the *minimal surface equation*, which is

$$\nabla \cdot \left(\frac{\nabla u}{\sqrt{1 + |\nabla u|^2}}\right) = 0 \tag{4}$$

Locally, ∇u is a constant, and so eq. (4) looks like $\Delta u = 0$, and so solutions have similar local properties. But the existence theory for Laplace's equation is 'trivial', whereas the existence theory for eq. (4) may fail. That is, global properties are important.

For entire solutions (i.e. solutions defined all of $\mathbb{R}^n$), global behaviour is very different.

Theorem 0.4 (Liouville). If $u : \mathbb{R}^n \to \mathbb{R}$ is C^2, $\Delta u = 0$ and u is bounded, then u is a constant.

Theorem 0.5 (Bernstein). (The only entire solutions to eq. (4) in $\mathbb{R}^n$ are planar (i.e. u is linear)) if and only if $n \leq 7$.

1 Harmonic functions

1.1 Basic properties

Let $\Omega \subseteq \mathbb{R}^n$ be a domain, i.e. open and connected.

Definition 1.1 (hamonic, subharmonic, superharmonic)
A function $u \in C^2(\Omega)$ is *harmonic* if $\Delta u = 0$. It is *subharmonic* if $\Delta u \geq 0$ and *superharmonic* if $\Delta u \leq 0$.

Let $B_\rho(y)$ denote the open ball with centre y and radius ρ. Then

Theorem 1.2 (mean value property). If $u \in C^2(\Omega)$ is subharmonic, and $B_r(y) \subseteq \Omega$, then

$$u(y) \leq \frac{1}{\omega_n r^n} \int_{B_r(y)} u(x)\mathrm{d}x \tag{5}$$

where $\omega_n = |B_1(0)|$. Moreover,

$$u(y) \leq \frac{1}{n\omega_n r^{n-1}} \int_{\partial B_r(y)} u(x)\mathrm{d}x \tag{6}$$

If u is superharmonic, then the inequalities are reversed. If u is harmonic, then equality holds.

Proof. We have that

$$\begin{aligned} 0 &\leq \int_{B_\rho(y)} \Delta u \mathrm{d}x \\ \text{integrating by parts} &= \int_{\partial B_\rho(y)} \nabla u \cdot w \mathrm{d}x \text{ where } w \text{ is the outwards normal} \\ &= \rho^{n-1} \int_{S^{n-1}} w \cdot \nabla u(y + \rho w)\mathrm{d}w \\ &= \rho^{n-1} \int_{S^{n-1}} \frac{\partial}{\partial \rho} u(y + \rho w)\mathrm{d}w \end{aligned}$$

where we use the fact that $\rho w = x - y$. Exchanging integrals and derivatives,

$$0 \leq \frac{\partial}{\partial \rho} \int_{S^{n-1}} u(y + \rho w)\mathrm{d}w$$

Thus, the map

$$\rho \mapsto \int_{S^{n-1}} u(y + \rho w)\mathrm{d}w$$

is increasing. Thus,

$$\int_{S^{n-1}} u(y + \rho w)\mathrm{d}w \leq \int_{S^{n-1}} y(y + r w)\mathrm{d}w$$

for $0 \leq \rho \leq r$. Taking the limit as $\rho \to 0$, we get eq. (6). Integrating in r to get eq. (5). The superharmonic case is similar. The harmonic case follows from the subharmonic and superharmonic cases. □

Remark 1.3. The mean value property characterises harmonic functions. See examples sheet 1.

Lecture 2

Theorem 1.4 (strong maximum principle)**.** Suppose $u \in C^2(\Omega)$ is subharmonic on Ω. Suppose there exists $y_0 \in \Omega$ such that

$$u(y_0) = \sup_\Omega u$$

Then u is constant.

Remark 1.5. If u is superharmonic, then we have a corresponding statement for when $u(y_0) = \inf_\Omega u$. If u is harmonic, then either sup or inf work.

Proof. Let $M = \sup_\Omega u < \infty$, and

$$\Sigma = \{y \in \Omega \mid u(y) = M\}$$

By assumption, Σ is non-empty, as $y_0 \in \Sigma$. Since u is continuous, Σ is closed. Since Ω is connected, it suffices to show that Σ is open, since this implies $\Sigma = \Omega$.

Pick $y \in \Sigma$. By the mean value property, for $\rho > 0$ such that $\overline{B_\rho(y)} \subseteq \Omega$, we have that

$$M = u(y) \leq \frac{1}{\omega_n \rho^n} \int_{B_\rho(y)} u(x) \mathrm{d}x$$

So

$$\int_{B_\rho(y)} (M - u(x)) \mathrm{d}x \leq 0$$

But $M - u(x) \geq 0$, and so it must be identically zero, i.e. $u \equiv M$ on $B_\rho(y)$. Hence $B_\rho(y) \subseteq \Sigma$, and Σ is open. □

Here, the strong maximum principle is easy as we have the mean value property. For more general elliptic PDEs, this is not the case. We will prove a weaker statement which does generalise.

Theorem 1.6 (weak maximum principle)**.** Suppose $\Omega \subseteq \mathbb{R}^n$ is a bounded domain, and $u \in C^2(\Omega) \cap C^0(\overline{\Omega})$. If u is subharmonic on Ω, then

$$\sup_\Omega u = \sup_{\partial\Omega} u$$

Proof. This follows from the strong maximum principle. Since Ω is bounded, $\sup_\Omega u$ is attained in $\overline{\Omega}$. So by the maximum principle, these cannot be attained in Ω unless u is constant. □

Remark 1.7. If u is superharmonic, we replace sup with inf, and if u is harmonic then both hold.

The mean value property states that u always an average of itself. In particular, this suggest that u cannot vary too much. Can we use this to relate $\sup u$ and $\inf u$?

Theorem 1.8 (Harnack's inequality)**.** Suppose $u \in C^2(\Omega)$, $u \geq 0$ and $\Delta u = 0$ in Ω. Then if $\Omega' \Subset \Omega$ is any bounded subdomain, we have

$$\sup_{\Omega'} u \leq C \inf_{\Omega'} u$$

where $C = C(n, \Omega, \Omega')$ does not depend on u.

Proof. First, choose $y \in \Omega$ and $\rho > 0$, such that $\overline{B_{4\rho}(y)} \subseteq \Omega$. Choose $x_1, x_2 \in B_\rho(y)$. By the mean value property,

$$u(x_1) = \frac{1}{\omega_n \rho^n} \int_{B_\rho(x_1)} u \leq \frac{1}{\omega_n \rho^n} \int_{B_{2\rho}(y)} u$$

On the other hand,

$$u(x_2) = \frac{1}{\omega_n (3\rho)^n} \int_{B_{3\rho}(y)} u \geq \frac{1}{\omega_n (3\rho)^n} \int_{B_{2\rho(y)}} u$$

Combining these, we see that

$$u(x_1) \leq 3^n u(x_2)$$

for all $x_1, x_2 \in B_\rho(y)$. So Harnack holds locally in balls, with constant independent of u. It is also independent of ρ, y as long as ρ is sufficiently small. Now choose x_1, x_2 in $\overline{\Omega'} \subseteq \Omega$, such that

$$\sup_{\Omega'} u = u(x_1) \quad \text{and} \quad \inf_{\Omega'} u = u(x_2)$$

By path connectedness of Ω, there exists a continuous map $\gamma : [0,1] \subseteq \overline{\Omega'}$, with $\gamma(0) = x_1$ and $\gamma(1) = x_2$. Choose ρ such that $4\rho < d(\gamma, \partial\Omega)$, and $N = N(\Omega', \Omega)$ such that we can cover γ by N-balls of radius ρ^2.

Apply the local result to each ball, and we get

$$u(x_1) \leq (3^n)^N u(x_2) = 3^{nN} u(x_2)$$

□

Theorem 1.9 (derivative estimates)**.** Suppose $u \in C^3(\Omega)$ is harmonic on Ω. Then if $\overline{B_\rho(y)} \subseteq \Omega$, we have that

$$|\mathrm{D}u(y)| \leq \frac{C}{\rho} \sup_{\partial B_\rho(y)} |u|$$

where $C = C(n)$.

Proof. Since $\Delta u = 0$,

$$0 = \mathrm{D}_i(\Delta u) = \Delta(\mathrm{D}_i u)$$

So $\mathrm{D}_i u$ is harmonic. By the mean value property,

$$\begin{aligned}
\mathrm{D}_i u(y) &= \frac{1}{\omega_n \rho^n} \int_{B_\rho(y)} u \\
&= \frac{1}{\omega_n \rho^n} \int_{B_\rho(y)} \nabla \cdot (0, \dots, u, \dots, 0) \mathrm{d}x \\
&= \frac{1}{\omega_n \rho^n} \int_{\partial B_\rho(y)} u(x) \cdot \nu_i(x) \mathrm{d}x
\end{aligned}$$

where $\nu(x)$ is the unit normal at x. But $|\nu_i(x)| \leq 1$, and so

$$|\mathrm{D}_i u(y)| \leq \frac{1}{\omega_n \rho^n} \sup_{\partial B_\rho(y)} |u| \int_{\partial B_\rho(y)} \mathrm{d}x = \frac{n}{\rho} \sup_{\partial B_\rho(y)} |u|$$

□

Remark 1.10. We can apply this result repeatedly, to get that for $\Omega' \subseteq \Omega'' \subseteq \Omega$, and any multi-index α, if $u \in C^{|\alpha|+2}(\Omega)$, with $\Delta u = 0$ in Ω, then

$$\sup_{\Omega''} |\mathrm{D}^\alpha u| \leq C \sup_{\Omega''} |u|$$

for some $C = C(n, \alpha, \Omega, \Omega')$.

That is,

$$\left\| \mathrm{D}^\alpha u \right\|_{L^\infty(\Omega')} \leq C \| u \|_{L^\infty(\Omega')}$$

By the mean value property, for some $y \in \overline{\Omega''} \subseteq \Omega$,

$$\sup_{\Omega''} u = |u(y)| = \left| c \int_{B_\rho(y)} u(x) \mathrm{d}x \right| \leq C \int_\Omega |u|$$

So

$$\left\| \mathrm{D}^\alpha(u) \right\|_{L^\infty(\Omega')} \leq C \| u \|_{L^1(\Omega)}$$

[2] This follows from the fact that Ω' is relatively compact

Theorem 1.11 (uniqueness of solutions for the Dirichlet problem)**.** Suppose Ω is bounded, and $u_1, u_2 \in C^2(\Omega) \cap C^0(\overline{\Omega})$ with

$$\begin{cases} \Delta u_1 = \Delta u_2 & \text{in } \Omega \\ u_1 = u_2 & \text{on } \partial\Omega \end{cases}$$

Then $u_1 = u_2$ in $\overline{\Omega}$.

Proof. Set $w = u_2 - u_2$. Then w is harmonic function in Ω, and $w = 0$ on $\partial\Omega$. By the weak maximum principle, we get that $w = 0$ in Ω. □

Remark 1.12. Of course, we can integrate by parts to get the same result, but the weak maximum principle will apply for non divergence form equations.

Lecture 3

Theorem 1.13 (Liouville)**.** If $u \in C^\infty(\mathbb{R}^n)$ is harmonic, and grows sublinearly at infinity. Then u is constant.

Remark 1.14. "growing sublinearly" means that

$$|u(x)| \leq C\left(1 + |x|^\alpha\right)$$

where $0 < \alpha < 1$.

Proof. From derivative estimates, we know that for all $x \in \mathbb{R}^n$, we have that

$$|Du(y)| \leq \frac{C}{\rho} \sup_{B_\rho(y)} |u|$$

Plugging in the growth assumption, we get that

$$|Du(y)| \leq \frac{C}{\rho}(1 + (\rho + |y|)^\alpha)$$

Taking $\rho \to \infty$, we get that $Du(y) = 0$. But y was arbitrary, and so we are done. □

1.2 Existence theory for harmonic functions

We will consider the classical problem of solving the Dirichlet problem for the Laplacian on bounded domains $\Omega \subseteq \mathbb{R}^n$, and $\varphi : \overline{\Omega} \to \mathbb{R}$ continuous. We wish to find $u \in C^\infty(\Omega) \cap C^0(\overline{\Omega})$ such that

$$\begin{cases} \Delta u = 0 & \text{in } \Omega \\ u = \varphi & \text{on } \partial\Omega \end{cases}$$

We will assume for simplicity that $\partial\Omega$ is smooth, and $\varphi \in C^\infty(\Omega)$.

We have (at least) three methods to solve this problem.

1. Hilbert space methods (c.f. Analysis of PDEs). Use the Riesz representation theorem to obtain a solution $u \in H^1(\Omega)$, and deal with regularity afterwards. This relies on the equation being linear.
2. Direct method of calculus of variations. We can rephrase $\Delta u = 0$ as a variational problem. That is, the Euler-Lagrange equation of
$$\int |Du|^2$$
and prove existence using the functional.
3. Perron's method. Use the fact that solvability in balls impies solvability in more general domains. This method is based on maximum principles.

Remark 1.15. In all cases, we obtain a rougher solution first, and improve regulaity later.

We will focus on the second method. Define

$$\mathscr{S} = \left\{ w \in \mathrm{H}^1(\Omega) \mid w - \varphi \in \mathrm{H}^1_0(\Omega) \right\}$$

That is, H^1 functions which agree with φ on the boundary. Clearly $\varphi \in \mathscr{S}$, and so $\mathscr{S}$ is non-empty. Set

$$\mathcal{E}[w] = \int_\Omega |\mathrm{D}w|^2$$

and define

$$\beta = \inf_{w \in \mathscr{S}} \mathcal{E}[w]$$

By the definition of the infimum, there exists a seqeunce $(w_j) \subseteq \mathscr{S}$, such that

$$\mathcal{E}[w_j] \to \beta$$

We want to extract a convergent subsequence and show that its limit is a solution. Clearly for j large, we have that

$$\int_\Omega \left|\mathrm{D}w_j\right|^2 \leq \beta + 1$$

Since $w_j - \varphi \in \mathrm{H}^1_0(\Omega)$, by the Poincaré inequality,

$$\int_\Omega \left|w_j - \varphi\right|^2 \leq C \int_\Omega \left|\mathrm{D}(w_j - \varphi)\right|^2$$

In particular, this implies that

$$\left\|w_j\right\|^2_{L^2(\Omega)} \leq C(\Omega, \varphi, \beta) < \infty$$

Indeed,

$$\left\|w_j\right\|^2_{L^2(\Omega)} \leq C\left\|\mathrm{D}w_j - \mathrm{D}\varphi\right\|^2_{L^2(\Omega)} \leq C(\Omega, \varphi, \beta)$$

Expanding the left hand side, we get

$$\left\|w_j\right\|^2_{L^2(\Omega)} - 2\left\langle w_j, \varphi \right\rangle_{L^2(\Omega)} \leq C(\Omega, \varphi, \beta)$$

In particular, by Young's inequality,

$$\left\|w_j\right\|^2_{L^2(\Omega)} \leq C(\varphi, \Omega, \beta) + \varepsilon\left\|w_j\right\|^2_{L^2(\Omega)} + \frac{1}{\varepsilon}\left\|\varphi\right\|^2_{L^2(\Omega)}$$

Take $0 < \varepsilon < 1$, and rearrange.

So we have that $\left\|w_j\right\|^2_{\mathrm{H}^1(\Omega)} \leq C$, so by Banach-Alaoglu,

$$w_{j_k} \rightharpoonup w \text{ in } \mathrm{H}^1(\Omega)$$

By Rellich-Kondrachov[3],

$$w_{j_k} \to w \text{ in } L^2(\Omega)$$

[3] Recall

Theorem 1.16 (Rellich-Kondrachov). Let Ω be bounded, $1 \leq p < n$, then

$$W^{1,p}(\Omega) \hookrightarrow L^{p^*}(\Omega)$$

and

$$W^{1,p}(\Omega) \Subset L^q(\Omega)$$

for $1 \leq q < p^*$. Here, p^* is the conjugate

$$p^* = \frac{np}{n-p}$$

In particular, when $p = 2$, $p^* > 2$ if $n > 2$.

for some $w \in H^1(\Omega)$. Hence for all $v \in H^1(\Omega)$, we have that

$$\int_\Omega Dw_j \cdot Dv \to \int_\Omega Dw \cdot Dv$$

Also, clearly we have that $w_{j_k} - \varphi \rightharpoonup w - \varphi$ in $H^1(\Omega)$, as φ is smooth. But $w_{j_k} - \varphi \in H^1_0(\Omega)$, but $H^1_0(\Omega) \subseteq H^1(\Omega)$ is norm closed, so it is weakly closed[4]. Hence $w - \varphi \in H^1_0(\Omega)$, so $w \in \mathscr{S}$. Finally, since $\mathcal{E}[\cdot]$ is *sequentially weakly lower semicontinuous*[5] in $H^1(\Omega)$, we have that

$$\mathcal{E}[w] \leq \liminf \mathcal{E}[w_{j_k}] = \beta$$

Hence $\mathcal{E}[w] = \beta$. We have found a global minimum w. That is, for all $v \in H^1_0(\Omega)$, $w + tv \in \mathscr{S}$,

$$f(t) = \mathcal{E}[w + tv] \geq \mathcal{E}[w]$$

Thus, $f'(0) = 0$. But

$$\begin{aligned} f'(0) &= D\mathcal{E}[w](v) \\ &= \lim_{t \to 0} \frac{\mathcal{E}[w + tv] - \mathcal{E}[w]}{t} \\ &= 2 \int_\Omega Dw \cdot Dv \end{aligned}$$

In particular,

$$\int_\Omega Dw \cdot Dv = 0$$

is the weak formulation of $\Delta w = 0$. We will next upgrade the regularity, and show that weak solutions are in fact smooth.

1.3 Interior regularity

We wish to improve the regularity of the weak solution. What we have shown is that there exists a weak solution $u \in L^1(\Omega)$, such that

$$\int_\Omega u \Delta v = 0$$

for all $v \in C^\infty_c(\Omega)$.

[4] This follows from Hahn-Banach. In fact, it follows for any convex subset of a Banach space.

> **Lemma 1.17.** Let X be a Banach space, for $C \subseteq X$ convex, C is norm-closed if and only if C is weakly closed.

Proof. Weak closed implies norm closed is clear. For the converse, we can show that $X \setminus C$ is weakly open. Let $x_0 \in X \setminus C$, by the Hahn-Banach separation, there exists $\phi \in X'$ such that

$$\varphi|_C = 0 \quad \text{and} \quad \phi(x_0) \neq 0$$

Then

$$\left\{ x \in X \;\middle|\; |\phi(x)| > \frac{1}{2}|\phi(x_0)| \right\} \subseteq X \setminus C$$

is a weakly open neighbourhood of x_0. □

[5] That is, if $u_j \rightharpoonup u$ in $H^1(\Omega)$, then

$$\mathcal{E}[u] \leq \liminf \mathcal{E}[u_j]$$

To see this, note that

$$\int_\Omega Du_j \cdot Dv \to \int_\Omega Du \cdot Dv$$

Setting $u = v$, we see that

$$\int_\Omega Du_j \cdot Du \to \int_\Omega |Du|^2$$

Thus,

$$\begin{aligned} \mathcal{E}[u] &= \lim \int_\Omega Du_j \cdot Du \\ &= \liminf \int_\Omega Du_j \cdot Du \\ &\leq \liminf \mathcal{E}[u_j]^{1/2} \mathcal{E}[u]^{1/2} \end{aligned}$$

Theorem 1.18 (Weyl's lemma)**.** Weakly harmonic functions are smooth. That is, for $\Omega \subseteq \mathbb{R}^n$ a domain, $u \in L^1_{\text{loc.}}(\Omega)$, if we have u is a weak solution to Laplace's equation, then u is C^∞ and $\Delta u = 0$ in Ω.

Proof. Mollify u. Take $\varphi \in C^\infty(\mathbb{R}^n)$, such that

- $0 \leq \varphi$,
- $\varphi(x) = 0$ for $|x| \geq 1$,
- $\int_{\mathbb{R}^n} \varphi = 1$,
- φ is radially symmetric[6].

For $\sigma > 0$, set

$$\varphi_\sigma(x) = \frac{1}{\sigma^n} \varphi\left(\frac{x}{\sigma}\right)$$

Then $\varphi_\sigma \in C_c^\infty(B_\sigma(0))$ is nonnegative and has integral 1.

Define

$$u_\sigma(x) = (\varphi_\sigma * u)(x)$$

This is well defined for

$$x \in \Omega_\sigma = \{x \in \Omega \mid d(x, \partial\Omega) \geq \sigma\}$$

Then u_σ is smooth, $u_\sigma \to u$ in $L^1_{\text{loc.}}(\Omega)$. Moreover, $\Delta u_\sigma = 0$. To see this,

$$\frac{\partial}{\partial x^i} u_\sigma = \int_\Omega u(y) \frac{\partial}{\partial x^i} \varphi_\sigma(x - y) \mathrm{d}y = -\int_\Omega u(y) \frac{\partial}{\partial y^i} \varphi_\sigma(x - y) \mathrm{d}y$$

and so

$$\Delta_x u_\sigma(x) = \int_\Omega u(y) \Delta_y \varphi_\sigma(x - y) \mathrm{d}y = 0$$

as u is weakly harmonic.

By the a priori derivative estimates for harmonic functions, for $\Omega' \Subset \Omega$,

$$\sup_{\Omega'} |\mathrm{D}^\alpha u_\sigma| \leq C \int_{\Omega'_{\sigma_1}} |u_\sigma|$$

for some $\sigma_1(\Omega')$ small, where

$$\Omega'_\sigma = \Omega' \cup \{x \in \Omega \mid d(x, \partial\Omega') < \sigma\}$$

Since $u_\sigma \to u$ in $L^1_{\text{loc.}}(\Omega)$, for σ small enough,

$$C \int_{\Omega'_{\sigma_1}} |u_\sigma| \leq C \int_{\Omega'_{\sigma'}} (|u| + 1)$$

Hence

$$\sup_{\Omega'} |\mathrm{D}^\alpha u_\sigma| \leq C \int_{\Omega'_{\sigma_1}} (|u| + 1)$$

So $\mathrm{D}^\alpha u_\sigma$ is uniformly bounded in $L^\infty(\Omega')$. Hence (as bounded derivatives imply equicontinuity) by Arzela-Ascoli, there exists a subsequence (σ_j) such that $\sigma_j \to 0$, and there exists $\widetilde{u} \in C^\infty(\Omega)$ such that $u_{\sigma_j} \to \widetilde{u}$ in $C^k(\Omega')$ for all k. Hence

$$\Delta \widetilde{u} = \lim_{j \to \infty} \Delta u_{\sigma_j} = 0$$

in Ω, as Ω' was arbitrary. By properties of mollifiers, $u_\sigma \to u$ a.e. in Ω, and so $\widetilde{u} = u$ a.e. □

Remark 1.19. We do not say anything about boundary regularity. It is possible to get (at least) $u \in C^0(\overline{\Omega})$.

Let's now improve our C^∞ existence result to C^0.

[6] This is not a standard assumption for mollifiers, but we can assume this.

Theorem 1.20 (existence and uniqueness for the Dirichlet problem with C^0 data)**.** Suppose Ω is bounded with $\partial\Omega$ sufficiently regular, then for any $\varphi \in C^0(\partial\Omega)$, there exists a unique $u \in C^\infty(\Omega) \cap C^0(\overline{\Omega})$ solving

$$\begin{cases} \Delta u = 0 & \text{in } \Omega \\ u = \varphi & \text{on } \partial\Omega \end{cases}$$

Remark 1.21. We might have

$$\int_\Omega |Du|^2 = \infty$$

for this solution.

Proof. Choose a sequence $(\varphi_n) \subseteq C^\infty(\mathbb{R}^n)$, such that $\varphi_n \to \varphi$ on $\partial\Omega$, in L^∞. Then we know that there exists $u_n \in C^\infty(\Omega) \cap C^0(\overline{\Omega})$, such that

$$\begin{cases} \Delta u_n = 0 & \text{in } \Omega \\ u_n = \varphi_n & \text{on } \partial\Omega \end{cases}$$

Then for all $n, m \in \mathbb{N}$, $\Delta(u_n - u_m) = 0$ in Ω, and $u_n - u_m = \varphi_n - \varphi_m$ on $\partial\Omega$. By the weak maximum principle,

$$\sup_{\overline{\Omega}} |u_n - u_m| \le \sup_{\partial\Omega} |u_n - u_m| = \left\|\varphi_n - \varphi_m\right\|_{L^\infty} \to 0$$

as $n, m \to \infty$. So (u_n) is Cauchy in $C^0(\overline{\Omega})$, which is a Banach space, so by completeness, there exists $u \in C^0(\overline{\Omega})$ such that $u_n \to u$ uniformly on $\overline{\Omega}$. In particular, $u = \varphi$ on $\partial\Omega$.

By the derivative estimates, (u_n) converges in $C^k(\Omega')$, for any $\Omega' \Subset \Omega$, and so u is smooth in the interior. Thus, $\Delta u = 0$ in the interior. □

Remark 1.22. A sufficient condition for regularity is that $\partial\Omega$ is C^2. More generally, it is enough to have the *exterior sphere condition*, which says that for all $z \in \partial\Omega$, there exists $B_\rho(y)$, such that $\overline{B_\rho(y)} \cap \partial\Omega = \{z\}$. There exists bounded domains in which this fails, and the conclusion of the theorem fails in that case as well. For example, when the boundary has a cusp.

2 General second order elliptic operators

From now on, write

$$Lu = a^{ij}\partial_i\partial_j u + b^i\partial_i u + cu$$

We will work on $\Omega \subseteq \mathbb{R}^n$ open, $u \in C^2(\Omega)$, $a^{ij}, b^i, c : \Omega \to \mathbb{R}$ and consider the Dirichlet problem

$$\begin{cases} Lu = f & \text{in } \Omega \\ u = \varphi & \text{on } \partial\Omega \end{cases}$$

for given f and φ. If we can write L in divergence form

$$Lu = \partial_i(a^{ij}\partial_j u) + \widehat{b}^i\partial_i u + cu$$

then we can use Hilbert space methods, as in Analysis of PDEs. But if a^{ij} is only C^0 say, we will need Schauder theory.

The idea is to deform L into Δ using a series of rescalings. In particular, this does not involve Sobolev spaces. Since $u \in C^2(\Omega)$, we can assume a^{ij} is symmetric.

Definition 2.1

We say that L is *elliptic* in Ω if the matrix (a^{ij}) is positive definite in Ω. So that

$$0 \le \lambda(x)|\xi|^2 \le a^{ij}(x)\xi^i\xi^j \le \Lambda(x)\left|\xi^2\right|$$

for all $\xi \in \mathbb{R}^n$ non-zero. In particular, λ is the minimum eigenvalue, and Λ is the maximum.

L is *strictly elliptic* if there exists λ_0 such that $0 < \lambda_0 \le \lambda(x)$ for all x. L is *uniformly elliptic* if it is

elliptic, and $\Lambda(x)/\lambda(x)$ is uniformly bounded.

Remark 2.2. In general, uniformly elliptic does not imply strictly elliptic.

Example 2.3

The minimal surface equation

$$\nabla \cdot \left(\frac{Du}{\sqrt{1+|Du|^2}} \right) = 0$$

has

$$a^{ij} = \left(\delta_{ij} - \frac{D_i u D_j u}{1+|Du|^2} \right) \frac{1}{\sqrt{1+|Du|^2}}$$

This is elliptic but not uniformly elliptic.

Lecture 4

We are interested in general second order elliptic operators, with $a^{ij}, b^i, c \in C^{0,\alpha}(\Omega)$. In particular, we cannot write them in divergence form, as a^{ij} are not C^1.

2.1 Basic properties

Theorem 2.4 (weak maximum principle)**.** Suppose that L is elliptic and that

$$\sup_\Omega \left| \frac{b^i}{\lambda} \right| < \infty \tag{(*)}$$

for some i. Moreover, suppose Ω is bounded, open, $u \in C^2(\Omega) \cap C^0(\overline{\Omega})$, such that $Lu \geq 0$[a].

Then

- if $c = 0$, then
$$\sup_\Omega u = \sup_{\partial\Omega} u$$
- if $c \leq 0$, then
$$\sup_\Omega u \leq \sup_{\partial\Omega} u^+$$

where $u^+ = \max(u, 0)$.

[a] We say that u is a *subsolution*.

Remark 2.5. The assumption that $c \leq 0$ is crucial. For example, when $n = 1$, let $\Omega = (0, \pi)$, and $u'' + u = 0$. Then $u(x) = \sin(x)$ is a solution. But then $\sup_\Omega u = 1$, and $\sup_{\partial\Omega} u^+ = 0$.

For $n = 2$, $\Omega = (0, \pi)^2$, and $\Delta u + 2u = 0$. Then

$$u(x, y) = \sin(x)\sin(y)$$

has the same properties as the above.

Proof. First suppose $c = 0$. If $Lu > 0$ in Ω, then in fact the strong maximum principle holds. Indeed, if $x_0 \in \Omega$ is a local maximum, then

$$\partial_i u(x_0) = 0 \quad \text{and} \quad \partial_i \partial_j u(x_0) \preccurlyeq 0$$

Since $a^{ij}(x_0) \succcurlyeq 0$, we have that

$$a^{ij}\partial_i \partial_j u(x_0) = \operatorname{tr}\left(A \nabla^2 u(x_0) \right) \leq 0$$

Hence $0 < Lu(x_0) = a^{ij}\partial_i\partial_j u(x_0) + b^i \partial_i u(x_0) \leq 0$. Contradiction. More generally, if $Lu \geq 0$ in Ω, consider

$$v(x) = e^{\gamma x_1}$$

for some $\gamma > 0$ to be chosen. Here, we assume without loss of generality that $(*)$ holds for $i = 1$. Then

$$\partial_1 v = \gamma e^{\gamma x_1} \quad \text{and} \quad \partial_i v = 0 \text{ for } i \neq 1$$

and

$$\partial_1 \partial_1 v \gamma^2 e^{\gamma x_1}$$

and all other second derivatives are zero. So

$$\begin{aligned} Lv &= e^{\gamma x_1}(a^{11}\gamma^2 + b^1\gamma) \\ &\geq e^{\gamma x_1}(\lambda\gamma^2 + b^1\gamma) \\ &= \gamma e^{\gamma x_1}\left(\gamma^2 + \frac{b^1}{\lambda}\gamma\right) \end{aligned}$$

This is positive for γ large enough. Hence

$$L(u + \varepsilon v) > 0$$

for all $\varepsilon > 0$. By the above,

$$\begin{aligned} u(x) &\leq \sup_{\Omega}(u + \varepsilon v) \quad \text{as } v \geq 0 \\ &\leq \sup_{\partial\Omega}(u + \varepsilon v) \\ &\leq \sup_{\partial\Omega} u + \varepsilon \sup_{\partial\Omega} v \end{aligned}$$

Taking $\varepsilon \to 0$, we get that$u(x) \leq \sup_{\partial\Omega} u$. Since this is true for all $x \in \Omega$, then

$$\sup_{\Omega} u \leq \sup_{\partial\Omega} u$$

The reverse inequality is trivial.

Now suppose $c \leq 0$. Define

$$L_0 u = a^{ij}\partial_i\partial_j u + b^i\partial_i u$$

and consider

$$\Omega^+ = \{x \in \Omega \mid u(x) > 0\}$$

Since $cu \leq 0$ on Ω^+,

$$L_0 u = Lu - cu \geq 0$$

on Ω^+. Note if $\Omega^+ = \varnothing$, then $\sup_\Omega u \leq 0$, and $u^+ = 0$, and the conclusion is trivial. Thus, without loss of generality assume $\Omega^+ \neq \varnothing$. Then there exists $x_0 \in \partial\Omega \cap \partial\Omega^+$, with $u(x_0) \geq 0$. If not, then

$$\partial\Omega^+ \cap \partial\Omega = \varnothing$$

and so $\partial\Omega^+ \subseteq \Omega$, and so it $\partial\Omega^+ \subseteq \Omega \setminus \Omega^+$. So $u|_{\partial\Omega^+} \leq 0$. But this contradicts the first part for L_0 on Ω^+.

Hence

$$\begin{aligned} \sup_{\Omega} u &= \sup_{\Omega^+} u \\ &= \sup_{\partial\Omega^+} u \\ &\leq \sup_{\partial\Omega} u \\ &\leq \sup_{\partial\Omega} u^+ \end{aligned}$$

□

Corollary 2.6. Let Ω be bounded open, $u \in C^2(\Omega) \cap C^0(\overline{\Omega})$, L is elliptic, with the same bound on Ω, and $c \leq 0$ in Ω. Then

1. if $Lu \leq 0$ in Ω, then

$$\inf_{\Omega} u \geq \inf_{\partial\Omega} u^-$$

where $u^- = \min(u, 0)$.

2. if $Lu = 0$, then
$$\sup_{\partial\Omega} |u| = \sup_{\Omega} |u|$$

Proof. Exercise. □

Corollary 2.7. Let L as above, and suppose we have $u, v, w \in C^2(\Omega) \cap C^0(\overline{\Omega})$, such that
$$Lu \geq 0 \quad Lv = 0 \quad \text{and} \quad Lw \leq 0$$
Then

1. if $u \leq v$ on $\partial\Omega$, then $u \leq v$ in $\overline{\Omega}$.
2. if $v \leq w$ on $\partial\Omega$, then $v \leq w$ in $\overline{\Omega}$.

Proof. Exercise. □

We want to build towards a strong maximum principle. For this, we need

Theorem 2.8 (Hopf boundary point lemma)**.** Let $\Omega \subseteq \mathbb{R}^n$ be open, and take $y \in \partial\Omega$. Suppose $\partial\Omega$ satisfies the *interior sphere condition*[a] at y. Let L be uniformly elliptic in Ω, with
$$\sup_{\Omega} \frac{|b|}{\lambda} + \sup_{\Omega} |c|\lambda < \infty$$
Suppose $u \in C^2(\Omega) \cap C^0(\{y\} \cup \Omega)$, such that $u(y) \geq u(x)$ for all $x \in \Omega$, and $Lu \geq 0$ in Ω. Finally, assume one of the following holds:

(i) $c = 0$ in Ω,

(ii) $c \leq 0$ in Ω and $u(y) \geq 0$,

(iii) $u(y) = 0$

Then
$$\frac{\partial u}{\partial \nu} > 0$$
if it exists, where ν is the outwards pointing normal at y to $\partial B_R(z)$, coming from the interior sphere condition.

[a] That is, there exists $R > 0$, $z \in \Omega$, such that $B_R(z) \subseteq \Omega$, $y \in \partial B_R(z)$.

Remark 2.9. The weak maximum principle implies that
$$\frac{\partial u}{\partial \nu} \geq 0$$
and so the content of the theorem is the strict inequality.

Proof. Let
$$A = B_R(z) \setminus B_r(z)$$
for some $0 < r < R$. We will first solve cases (i) and (ii). On A, consider
$$v(x) = e^{-\alpha|x-z|^2} - e^{-\alpha R^2}$$
First note that on A, $v > 0$.
$$\partial_i v(x) = -2\alpha(x_i - z_i)e^{-\alpha|x-z|^2}$$
and
$$\partial_i\partial_j v(x) = -2\alpha\delta_{ij}e^{-\alpha|x-z|^2} + 4\alpha^2(x_i - z_i)(x_j - z_j)e^{-\alpha|x-z|^2 s}$$

So on A,

$$Lv = e^{-\alpha|x-z|^2}(a^{ij}4\alpha^2(x_i - z_i)(x_j - z_j) - 2\alpha a^{ii} - 2\alpha b^i(x_i - z_i) + c) - ce^{-\alpha R^2}$$

By ellipticity and the sign of c,

$$\begin{aligned} Lv &\geq 2^{-\alpha|x-z|^2}(4\alpha^2\lambda(x)|x-z|^2 - 2\alpha n\Lambda(x) - 2\alpha|b||x-z| - |c|) \\ &\geq e^{-\alpha|x-z|^2}\lambda(x)\left(\alpha^2R^2 - 2\alpha n\sup_\Omega\frac{\Lambda}{\lambda} - \alpha R\sup_\Omega\frac{|b|}{\lambda} - \sup_\Omega\frac{|c|}{\lambda}\right) \end{aligned}$$

where r is chosen such that $|x-z|^2 \geq (R/2)^2$. In particular, this is positive for α large enough. Fix such an α. □